Architecture and Diagrams

This document presents a consistent set of three diagrams (system context, use cases and a draft component view) and the architectural description behind them. The diagrams use the same actor, module and requirement names as the requirements specification (05) and the architectural decisions (06). The component view is a draft of the target MVP structure, not an implemented system. The last section checks whether requirements, components, data and scenarios agree.

Architectural Drivers

Five groups of requirements shape the structure most. Independent assessment and deterministic correctness (FR04, FR07, NFR02) require the server to own answers and the hint prohibition, so Assessment is the source of truth and Tutor cannot change scores. AI hints must be optional and safe (FR06, NFR04, NFR06): the AI adapter is the only path to the provider, sends minimal data and falls back to templates. Family data needs strict access control (FR10, FR14, NFR01), enforced by middleware and by repositories that expose only linked children. Kazakh and Russian are both mandatory (FR16), so every item, hint and report needs two language variants. A two-person team and a reproducible local environment (NFR10) favour a modular monolith in one process with PostgreSQL.

System Context

Figure 1 shows MathVerity as one system with four external parties. Learners and parents use the browser client. Content is authored by the two project developers through a restricted internal capability (FR11). The AI provider is the only external system; PostgreSQL is internal and therefore not drawn here. The supervisor has no role in the system and no access to children’s data.

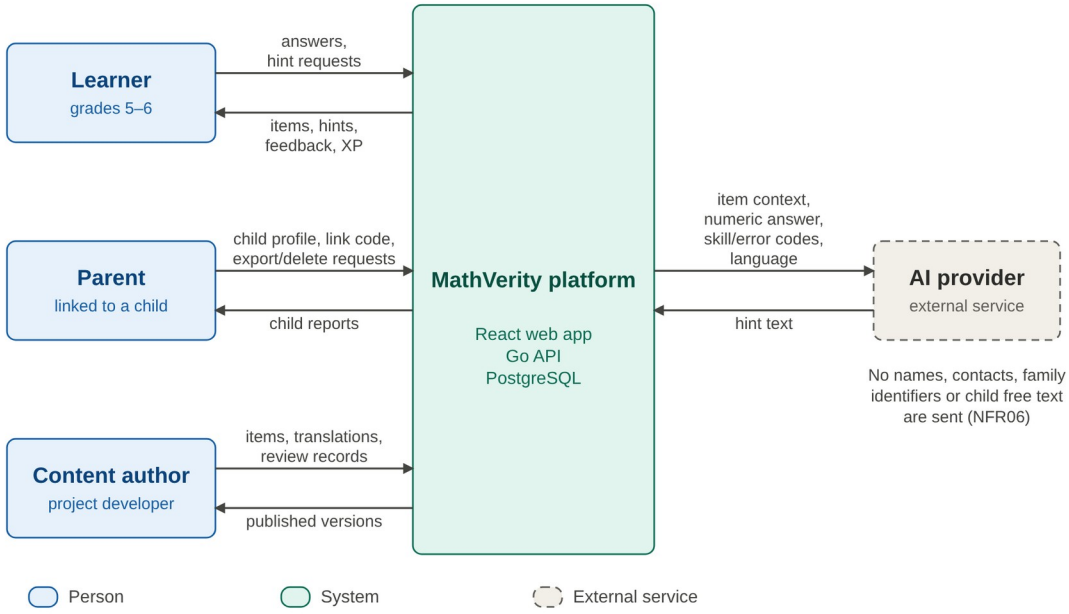


Figure 1. System context of MathVerity.

Table 1. External parties and interfaces

Party	Sends to MathVerity	Receives from MathVerity	Requirements
Learner	answers, hint requests, language choice	items, hints, feedback, XP, own progress	FR03–FR09, FR16
Parent	sign-in, child profile, link code, consent, export and deletion requests	child reports, link status	FR01, FR02, FR10, FR14
Content author (project developer)	items, translations, review records	published versions, template previews	FR11, FR16
AI provider	hint text, checked against a schema	item context, numeric answer, skill and error codes, requested language	FR06, NFR04, NFR06

Use Cases

Figure 2 groups thirteen use cases by actor. Signing in applies to every role; the content author’s link to UC01 is left out of the figure for readability. Dashed ellipses are Should requirements (FR12 and FR15). Hints, next-item suggestions and delayed reviews extend practice or independent assessment; the hint extension is not available in independent assessment (FR07). The AI provider is a secondary actor that takes part only in UC05.

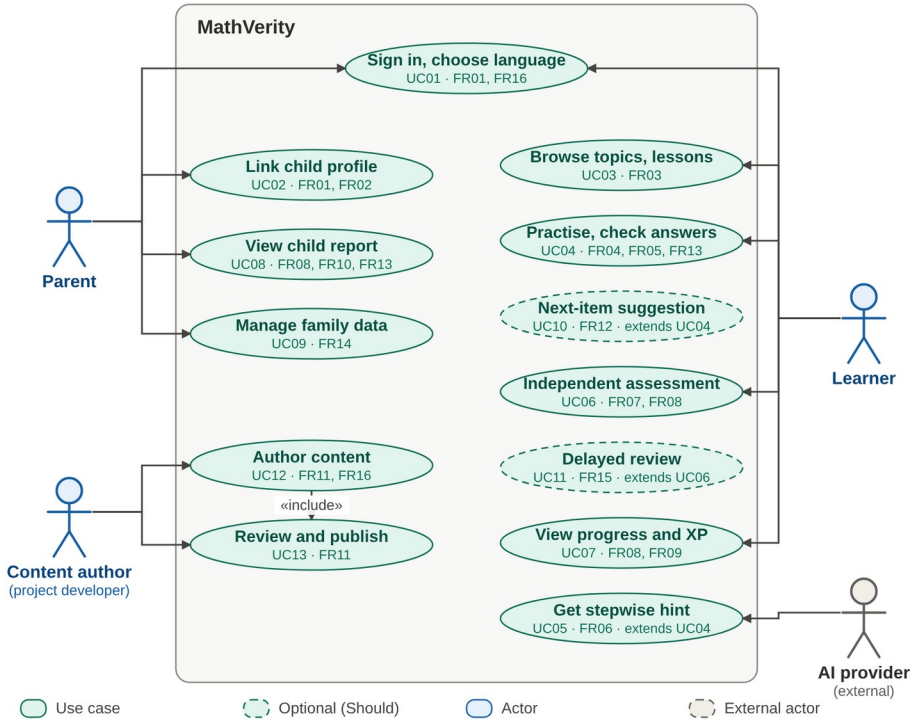


Figure 2. Use cases of MathVerity.

Table 2. Use cases and their requirements

ID	Use case	Actor	Requirements	Main components
UC01	Sign in, choose language	Learner, Parent; also the Content author	FR01, FR16	Identity/Family, client
UC02	Link child profile	Parent	FR01, FR02	Identity/Family
UC03	Browse topics, lessons	Learner	FR03	Content
UC04	Practise, check answers	Learner	FR04, FR05, FR13	Assessment, Content, Reporting
UC05	Get stepwise hint	Learner (AI provider as secondary actor)	FR06	Assessment, Tutor, AI adapter
UC06	Independent assessment	Learner	FR07, FR08	Assessment, Progress
UC07	View progress and XP	Learner	FR08, FR09	Progress, Gamification
UC08	View child report	Parent	FR08, FR10, FR13	Reporting, Identity/Family, Progress
UC09	Manage family data	Parent	FR14	Identity/Family
UC10	Next-item suggestion (Should)	Learner	FR12	Progress
UC11	Delayed review (Should)	Learner	FR15	Progress, Assessment
UC12	Author content	Content author	FR11, FR16	Content
UC13	Review and publish	Content author	FR11	Content

Component View (Draft)

Figure 3 is a draft. The React client is delivered as static files and calls the Go API over HTTPS on the same origin. One Go process holds the modules; arrows show the main calls. Only four internal repositories (Identity/Family, Content, Assessment and Progress/XP) access PostgreSQL, so Tutor and Reporting never touch the database directly. The daily backup is drawn dashed because NFR08 has no component design yet.

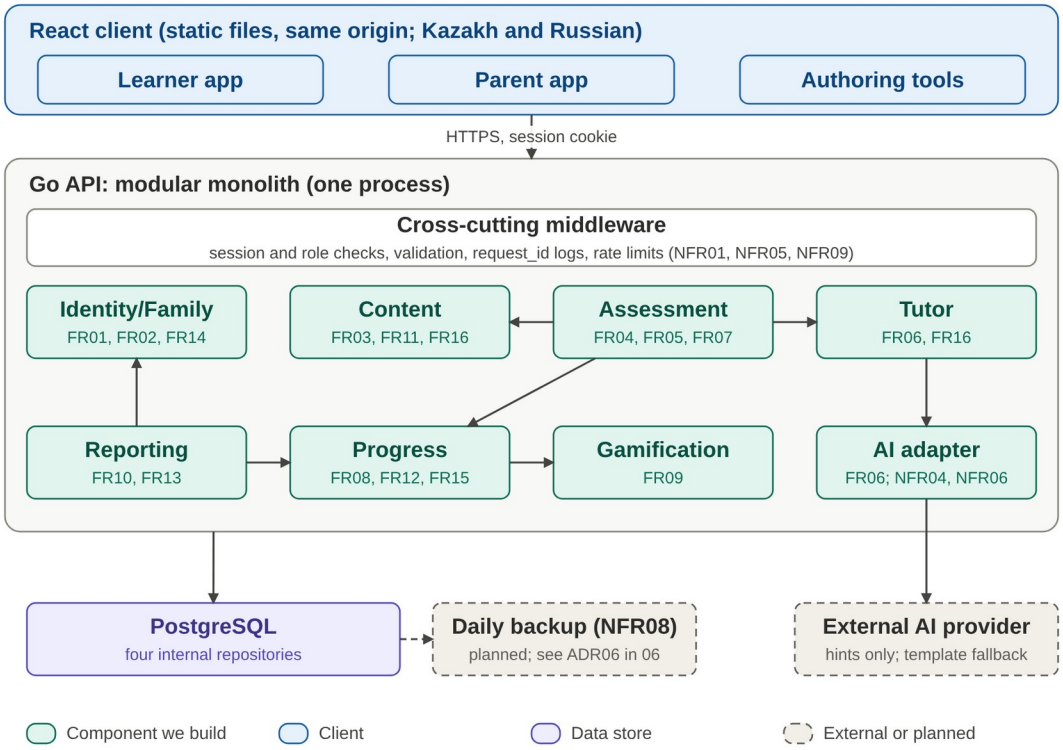


Figure 3. Draft component diagram. Requirement identifiers show which module serves which requirement.

Table 3. Component responsibilities

Component	Responsibility	Requirements	Calls
Identity/Family	accounts, roles, sessions, child profiles, one-time link codes, family data requests	FR01, FR02, FR14, NFR01	none
Content	items, lessons, language variants, versions, review records	FR03, FR11, FR16	none
Assessment	answer checking, error diagnosis, attempt mode, independent variants, hint prohibition	FR04, FR05, FR07, NFR02	Content, Tutor, Progress
Tutor	hint levels, template selection per language, safe hint response; no database access	FR06, FR16	AI adapter
AI adapter	minimal request, timeout, schema check and filtering, fallback signal	FR06, NFR04, NFR06	AI provider

Component	Responsibility	Requirements	Calls
Progress	skill statuses (rule 1.1), next-item selection, delayed review	FR08, FR12, FR15	Gamification
Gamification	XP and achievements, idempotent awards	FR09	none
Reporting	parent reports, visible activity, estimated platform time	FR10, FR13	Identity/Family, Progress
Middleware	session, role and object checks, validation, request_id, rate limits	NFR01, NFR05, NFR09	none
React client	learner, parent and authoring views in two languages	FR16, NFR07	Go API

Data Overview

Each entity in Table 4 has one owning module; other modules reach it through the owner. Personal data is limited to an account and a child alias. No names, contacts, family identifiers or free text from children are sent to the AI provider.

Table 4. Main data entities and owners

Entity	Owner	Main content	Requirements
Account and session	Identity/Family	user, role, server session	FR01, NFR01
Child profile	Identity/Family	alias, language, deletion state	FR01, FR14, FR16
Family link and code	Identity/Family	one-time code (15 minutes), link, revocation	FR02, FR14
Data request	Identity/Family	export or deletion request	FR14
Topic, lesson, skill	Content	four blocks, eight skills, published state	FR03
Item and versions	Content	templates, Kazakh and Russian variants, reference answer, diagnostic choices, hint templates	FR03, FR06, FR11, FR16
Review record	Content	author, reviewer, content version, timestamp	FR11
Attempt	Assessment	variant, mode, first answer, selected step, error code, hint use, idempotency key	FR04, FR05, FR06, FR07
Skill status	Progress	rule version, evidence, item count, date, status; review schedule	FR08, FR12, FR15
XP ledger, achievement	Progress/XP repository	awards, one per item instance	FR09
Activity window	Progress/XP repository (via Reporting)	visible-activity intervals, 60-second gap rule	FR13

Key Scenarios

The six scenarios in Table 5 follow the main paths through the components. They are the basis of the end-to-end tests in the verification plan.

Table 5. Key scenarios

ID	Scenario	Flow	Requirements	Tests
S1	Parent links a child	Parent signs in and picks a language; Identity/Family creates a session, a child profile and a one-time code (15 minutes); the learner uses the code and the link is recorded.	FR01, FR02, FR16	T01, T02, T26
S2	Practice with a hint	Content supplies a published item; Assessment checks the answer on the server and stores the selected step and error code; on a hint request Assessment calls Tutor, which asks the AI adapter and falls back to a template in the selected language if the call fails or is rejected; Progress may suggest the next item.	FR03–FR06, FR12	T03–T06, T12
S3	Independent assessment	Assessment creates a new variant, rejects hint requests with 403 and records the first answer once; Progress recalculates the status with rule 1.1; Gamification awards XP once per item instance; Progress schedules a delayed review.	FR07–FR09, FR15	T07–T09, T15
S4	Parent views a report	Reporting checks the family link, reads statuses and evidence from Progress and returns skills, assistance, independent results, recency, recommended review and estimated platform time; another family's child_id is refused.	FR08, FR10, FR13	T08, T10, T13, T16
S5	Content is authored and published	Developer A drafts an item with both language variants, reference answer, diagnostic steps and hints; developer B records a review; publishing creates a new version and earlier attempts keep theirs; self-approval and a missing language variant are rejected.	FR11, FR16	T11, T26
S6	Family data is deleted	Parent requests export or deletion; Identity/Family marks the profile deleted and active APIs stop returning it; backup deletion is documented separately.	FR14	T14

Alignment Check

Alignment was checked by tracing each functional requirement to a use case, a component, a data entity, a scenario and a test, and by checking each non-functional requirement against an architectural element. Table 6 gives the functional trace; coverage was verified from this table.

Table 6. Traceability of functional requirements

FR	Use case	Component	Data	Scenario	Test
FR01	UC01, UC02	Identity/Family; middleware	Account, Child profile	S1	T01

FR	Use case	Component	Data	Scenario	Test
FR02	UC02	Identity/Family	Family link and code	S1	T02
FR03	UC03	Content	Topic, lesson, skill; Item	S2	T03
FR04	UC04	Assessment; Content	Item; Attempt	S2	T04
FR05	UC04	Assessment	Attempt (step, error code)	S2	T05
FR06	UC05	Tutor; AI adapter; Assessment; Content	Hint event; Item templates	S2	T06
FR07	UC06	Assessment; Tutor	Attempt (mode, variant)	S3	T07
FR08	UC06, UC07, UC08	Progress	Skill status	S3, S4	T08
FR09	UC07	Gamification; Progress	XP ledger, Achievement	S3	T09
FR10	UC08	Reporting; Identity/Family; Progress	Family link; Skill status	S4	T10
FR11	UC12, UC13	Content	Item version; Review record	S5	T11
FR12	UC10	Progress	Skill status; Attempt history	S2	T12
FR13	UC04, UC08	Reporting	Activity window	S4	T13
FR14	UC09	Identity/Family	Data request; Child profile	S6	T14
FR15	UC11	Progress; Assessment	Review schedule	S3	T15
FR16	UC01, UC12	Content; Tutor; client	Item language variants; Child profile	S1, S5	T26

Result: all 16 functional requirements map to at least one use case, component, data entity, scenario and test. All 13 use cases and all eight modules are used by at least one requirement, and every functional requirement named in the use case table appears in the trace. Of the 10 non-functional requirements, 9 are addressed by an architectural element and 1 (NFR08) only by a decision, without a detailed design (Table 7).

Table 7. Coverage of non-functional requirements

ID	Area	Architectural element	Status
NFR01	Access control	Middleware role and object checks; Identity/Family; Reporting link check	Aligned
NFR02	Assessment reliability	Assessment evaluator with exact arithmetic (ADR02)	Aligned
NFR03	API latency	Single Go process and PostgreSQL; load profile in 05	Aligned
NFR04	AI failure	AI adapter timeout and Tutor template fallback	Aligned
NFR05	Input protection	Middleware validation; parameterised SQL	Aligned

ID	Area	Architectural element	Status
NFR06	Data minimisation	AI adapter boundaries in 06; language added to the request	Aligned
NFR07	Accessibility	React client (keyboard, contrast, 360 px); no server component	Aligned
NFR08	Recovery	Daily PostgreSQL backup decided in ADR06; no detailed design yet	Partly
NFR09	Observability	Middleware request_id and structured logs	Aligned
NFR10	Reproducibility	Docker Compose, pinned versions, static-hint mode (06)	Aligned

Table 8 lists the inconsistencies found between the documents and the proposed resolutions. Items 3 to 6 were corrected in the architectural decisions (06); items 1 and 2 need a change in the requirements specification (05).

Table 8. Findings and resolutions

No.	Finding	Resolution	Status
1	05 names Identity and Family as separate modules and lists UI as a module; 06 and the diagrams use one Identity/Family module and a React client.	Treat Identity and Family as one module; relabel FR01, FR02 and FR14 in 05 and map UI to the React client.	Open: edit 05
2	Test IDs T01–T26 in 05 use the same prefix as the technology sources T01–T15 cited in 04 and 06; NFR07 cites source T14 (WCAG) while T14 is also a test.	Rename tests to TC01–TC26 in 05.	Open: edit 05
3	FR13 (visible activity) is assigned to Reporting, which has no repository of its own in 06.	Activity windows are stored through the Progress/XP repository; stated in 06 and in Table 4.	Resolved
4	NFR08 (daily backup, 60-minute restore) has no component and no architectural decision.	ADR06 added in 06: daily PostgreSQL backup in the Docker environment, with a separate restore test. A detailed design is still needed.	Partly resolved
5	06 did not list the requested language among the data sent to the AI provider (NFR06).	Added to the AI adapter boundaries in 06.	Resolved
6	06 did not mention internal content authoring by the developers (FR11, access matrix).	Added to the context section of 06; the developers have no access to identifiable family results.	Resolved
7	Both languages are mandatory (FR16), but 06 did not state where the variants live.	Content stores both variants per item and hint template; the language travels with each request (Table 4).	Aligned